

Warm-up

(1) Set up (but don't compute!) an integral to compute the arc length of the curve given by $y = \sin(x^3)$ as x goes from 0 to 2π .

(2) Set up (but don't compute!) an integral for the arc length of the curve $y = 1 + 3x + e^x$ as y goes from 2 to $4 + e$.

(3) Set up (but don't compute!) an integral for the area of the surface obtained by taking the curve $x = \ln y^2 + 5y$ as y goes from 2 to 7 and rotating it around the y -axis.

Medium

(1) Compute the arc length of the curve $y = \frac{x^4}{8} + \frac{1}{4x^2}$ as x varies from 1 to 2.

(2) Compute the arc length of the curve $(y - 2)^3 = x^2$ between $y = 2$ and $y = 6$ for $x \geq 0$.

(3) Compute the area of the surface obtained by taking the curve $y = x^3$ as x goes from 0 to 1 and rotating it around the x -axis.

(4) Set up (but don't compute!) an integral for the area of the surface obtained by taking the curve $x^{2/3} + y^{2/3} = 1$ as x goes from 0 to 1 and rotating it around the x -axis.

Stretch

(1) Consider the shape obtained by taking the curve $y = 1/x$ for $x \geq 1$ and rotating it around the x -axis. Sketch a picture of this shape and compute its surface area. (How much paint would it take to coat this surface?)

(2) For the same shape from Problem (1), compute the volume it contains. (How much paint would it take to fill this shape? Can you square that with your answer to the previous problem?)